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Projects

ArXiv Research Intelligence Agent (Agentic RAG)

Jun 2026 - Aug 2026

<https://github.com/V1cky16112003/research-intelligence-agent>

- Built a full-stack QA system over 50,000 ArXiv ML papers: two-stage ingestion pipeline loading JSONL metadata into Neon Postgres, chunking abstracts and embedding with Nomic-embed-text-v2 (768-dim), stored in pgvector with HNSW indexing.
- Designed a four-node LangGraph agent (Planner, Executor, Critic, Reporter) with a Critic-driven retry loop – up to 3 Executor re-runs before forcing a final grounded answer with citations; LLM routing: Groq (Llama 3.3 70B) primary, Gemini 2.5 Flash fallback.
- Implemented RAGAS evaluation pipeline with 20-question golden set; CI gate blocks merges if faithfulness drops below 0.80 or answer relevancy below 0.75; metrics logged to MLflow on DagsHub.
- Deployed FastAPI backend on Hugging Face Spaces, React 18/Vite frontend on Vercel; GitHub Actions runs lint, tests, and RAGAS quality gate on every push.
- Stack: Python, FastAPI, LangGraph, pgvector, Groq, Gemini 2.5 Flash, React 18, Docker, MLflow, GitHub Actions

Chore Manager - Agentic Household Automation via MCP

Aug 2026 - Sep 2026

<https://github.com/V1cky16112003/chore-manager>

- Built a full-stack chore-management app with a custom MCP server exposing 10 tools (add/assign/complete chores, member management, round-robin logic), enabling direct LLM control from Claude Desktop/Web.
- Deployed the MCP server serverless on Vercel in stateless HTTP mode, resolving Starlette session-affinity and DNS-rebinding constraints incompatible with serverless.
- Implemented bearer-token auth via Starlette middleware with TransportSecuritySettings host allowlisting, hardening the endpoint against DNS-rebinding while staying reachable to the Claude connector.
- Diagnosed and resolved a chain of 5 production issues (404 routing, 421 invalid-host, 401 auth, SDK v1/v2 incompatibility) across 25 commits, taking the integration to a stable authenticated endpoint.
- Stack: Python, FastMCP, Starlette, Model Context Protocol (MCP), Vercel Serverless, Bearer-Token Auth, REST APIs

Multimodal Emotion Recognition on SEED-VII: A Reproducibility Critique of Reported

Jan 2026 - Aug 2026

EEG-Eye Benchmark and an Honest Leak-Free Baseline

<https://github.com/V1cky16112003/Kings-final-year-project>

- Audited MAET benchmark (IEEE TAFRC 2025, 71.28% claimed SEED-VII accuracy) by replicating the full architecture in PyTorch on KCL's A100 GPU cluster; proved the reported figure is an evaluation-leakage artefact unreachable under any leak-free protocol.
- Designed a block-size sweep across 20 subjects tracing a quantified leakage-accuracy curve that reproduced the original figure to within 0.17 pp at block size 44; confirmed via paired Wilcoxon tests ($W=0$, $p<2e-6$, Cohen's $d=3.72$).
- Benchmarked three honest methods – graph-convolutional model DAMS-GCT (42.34%), masked self-supervised pretraining (42.48%), and intensity-aware multitask MAET (47–48%) – with convergent nulls confirming the honest ceiling is feature-driven, not architecture-limited.
- Validated leakage on independent dataset (MultiPhysio-HRC, 44 participants): +13.81 pp inflation ($p<2e-8$, $d=1.115$), demonstrating leakage can manufacture an apparent classifier from a null dataset.
- Stack: Python, PyTorch, GNNs, Transformers, EEG Signal Processing, SLURM/HPC, scikit-learn, SciPy

Conversational Multimodal Image Recognition Chatbot

Jan 2025 - May 2025

<https://github.com/V1cky16112003/Conversational-Image-Recognition-Chatbot>

- Built a multimodal chatbot combining real-time image recognition (Gemini 2.0 Flash), OCR (Azure Computer Vision), and voice I/O (Azure Speech) with support for English, Tamil, Hindi, and Telugu.
- Designed a production REST API (FastAPI backend, React.js frontend) handling real-time multimodal inference across 4 languages with sub-second response latency.
- Stack: Gemini 2.0 Flash, Azure Cognitive Services, FastAPI, React.js, Python

Technical Skills

Programming: Python (proficient), SQL, JavaScript, Git

ML & AI: PyTorch, TensorFlow, Scikit-Learn, GNNs, Transformers, HuggingFace, Multimodal AI, NLP, Computer Vision

Signals & Modelling: EEG/EDA/BVP signal processing, Multimodal fusion, Uncertainty quantification, Transfer learning

Engineering & Backend: FastAPI, LangGraph, Docker, REST APIs, pgvector, Azure Cognitive Services, MySQL, MCP (Model Context Protocol)

Frontend & Deployment: React.js, Vite, Vercel, Hugging Face Spaces, GitHub Actions, MLflow

Education

King's College London

Sep 2025 - Sep 2026

MSc, Data Science

London, UK

- **Achievements:** Expected: Distinction, Sem 1 avg: 71.7%
- **Coursework:** Neural Networks & Deep Learning, Data Mining, Big Data Technologies, Databases & Data Warehousing, Statistics

SRM Institute of Science & Technology

Jun 2021 - May 2025

B.Tech, Computer Science and Engineering (GPA: 8.73/10)

Chennai, India

- **Achievements:** First Class with Distinction

Work Experience

Stealth Startup | *AI Security Researcher / Policy Architect*

Jun 2026 - Jul 2026

- Designed and stress-tested an authorization policy layer for Guard AI, a runtime security system that mediates every action an autonomous AI agent proposes, enabling it to allow, deny, transform, sandbox, or escalate actions before they reach the real world.
- Authored a Cedar-ready policy specification covering the OWASP Agentic Top 10 (ASI01–ASI10) and OWASP LLM Top 10, mapping five core risk types—data exfiltration, goal hijacking, memory poisoning, privilege escalation, and supply-chain compromise—to an action-by-resource decision matrix; this provided consistent risk assessment and streamlined compliance reviews
- Designed a core information-flow model – structural provenance tracking, a two-axis authorization framework, and a bounded declassification lattice – validated against six real production incidents including EchoLeak (CVE-2025-32711, CVSS 9.3) and Cursor Studio confused-deputy (CVE-2026-21520).
- Built and ran three adversarial stress-test passes, identifying and resolving 20+ internal contradictions and security gaps including a data-laundering path, a stale-authorization race condition, and a null-sentinel collision.

4i Apps Solutions Private Limited | *AI Intern (Clustering and Deep Learning)*

Dec 2023 - Jan 2024

- Built a K-Means segmentation pipeline on ~3–4 GB of purchase data, identifying customer behavioural patterns and translating findings into targeting recommendations for the business team.
- Trained and evaluated TensorFlow/Keras deep-learning models for image classification and NLP tasks, using FastAPI for inference serving and MLflow to log experiments; reported model performance that met project benchmarks, enabling the team to select the best model for production.

Data Science Intern

Jun 2023 - Jul 2023

- Built an end-to-end classification pipeline on commercial datasets, cleaning data and performing EDA with Pandas and NumPy, then developing and evaluating models with Scikit-Learn, which streamlined the workflow and reduced manual preprocessing effort for the team.
- Built dashboards in Matplotlib and Seaborn to present model results to clients, producing the charts and summaries used in business reporting.

Certifications

- Model Context Protocol: Advanced Topics: Anthropic · Jul 2026
- AI Fluency Framework & Foundations: Anthropic · Jul 2026